

DS	Session	Hashing	Question Information	Level 1	Challenge 98
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Question description

You are given an array A of length N which is initialised with 0 . You will be given Q queries of two types:

- 1 k : set value 1 at index k in array A
- 2 y : print the smallest index x which is greater than or equal to y and having value 1 . If there is no such index print -1 .

Note: Indexing is 1 based

Constraints

- $1 \leq n \leq 10^9$
- $1 \leq q \leq 5 * 10^5$
- $1 \leq y, k \leq n$

Input Format

First line contains two integers N and Q separated by a space.

The next Q lines contain the type of query (i.e. either a 1 or a 2), then a space, then for type 1 queries integer k and for type 2 queries integer y

Output Format

For each query type 2 , print in new line, the smallest index x which is greater than or equal to y and having value 1 . If there is no such index print -1 .

Explanation for Test case 1

- For first query: $2\ 3$, there is no index greater than or equal index 3 , having value -1 , so the answer is -1 .
- For second query: $1\ 2$, set value -1 at index 2 .
- For third query: $2\ 1$, index 2 is greater than index 1 , having value -1 , so the answer is 2 .
- For fourth query: $2\ 3$, there is no index greater than or equal index 3 , having value -1 , so the answer is -1 .
- For fifth query: $2\ 2$, index 2 is equal to index 2 , having value -1 , so the answer is 2 .

Logical Test Cases

Test Case 1

INPUT (STDIN)

5 5
2 3
1 2
2 1
2 3
2 2

EXPECTED OUTPUT

-1
2
-1
2

Test Case 2

INPUT (STDIN)

5 5
2 1
1 2
2 2
1 2
2 3

EXPECTED OUTPUT

-1
2
-1

Mandatory Test Cases

Test Case 1

KEYWORD

for(i=0;i<q;i++)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

225

Test Case 3

NLOC

38